

IT112: WEB SYSTEMS and TECHNOLOGIE'S

2nd Semester, AY: 2025-2026

Phase 7: Final Group Project

BentaBoard

"From bawat benta to tunay na kita—track it right."



Proposed By:

PeNiHAMiNo

BSIT 2A

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Presented to:

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IT112 Professor

Introduction

This final phase of our project marks the completion of BentaBoard, a business management system built to help small sellers track their daily sales, expenses, and profits. After working through six phases of development, from planning and backend setup to frontend design and connecting everything together, Phase 7 is all about making sure our system is truly ready for real use.

In this phase, we focused on four important tasks, testing every feature to catch and fix bugs, polishing the user interface so it looks clean and works smoothly on any device, deploying our project to a live server so anyone can access it online, and preparing proper documentation that explains how our system works. We also made sure our GitHub repository is organized and complete, with a clear README file that guides anyone who wants to understand or run our project.

This week was not about adding new features but about refining what we already built. We wanted to make sure that when we present BentaBoard, every button works, every form submits correctly, and every page looks professional. This phase represents the finishing touches that turn a school project into something we can be proud to show.

Objectives

At the end of this activity, we were able to:

1. Conduct functional and usability testing of a full-stack web system.
2. Polish and optimize their UI/UX for a better user experience.
3. Finalize and deploy their full project to a live server using Render or equivalent.
4. Write proper project documentation (README, installation/setup, features, contributors).
5. Prepare for defense by simulating demo presentation and Q&A.

I. System Overview & Objective

BentaBoard is a web-based business management system built to help small-scale sellers in the Philippines, such as sari-sari store owners, market vendors, and online sellers, keep track of their daily business operations. Many of these small business owners still use notebooks or mental math to record their sales and expenses, which often leads to mistakes and makes it hard to know if they are actually earning a profit. BentaBoard solves this problem by giving them a simple, easy-to-use online tool where they can record their products, sales, expenses, and purchases in one place. The system then automatically calculates their profit so they can see exactly how their business is doing.

The main goal of BentaBoard is to help small sellers move from manual tracking to a digital system without needing to learn complicated accounting software. Everything is designed to be straightforward, with clear buttons, simple forms, and easy-to-read reports. Whether the user is on a computer or a mobile phone, the system adjusts to fit their screen so they can manage their business anytime and anywhere.

II. Features implemented (CRUD, search, UI)

CRUD Operations (Create, Read, Update, Delete)

Our system fully supports all four CRUD operations across every major feature.

Products

- **Create:** Add new products with name, price, category, and stock level.
- **Read:** View all products in a table with stock indicators.
- **Update:** Edit product details like price or stock through a modal form.
- **Delete:** Remove products from inventory with a confirmation prompt.

Sales

- **Create:** Record a new sale by selecting a product and entering quantity. The total is automatically calculated.
- **Read:** View the complete sales history sorted by date, with the most recent first.
- **Update:** Sales records cannot be edited after submission to maintain accuracy of financial records.
- **Delete:** Sales records are deleted when a user wipes all their data.

Expenses

- **Create:** Log new expenses with category, description, and amount.
- **Read:** View all expenses in a table with red-colored amounts to show money going out.
- **Update:** Expenses can be edited if a mistake was made in the entry.
- **Delete:** Expenses are removed when data is wiped.

Purchases

- **Create:** Record inventory purchases from suppliers with supplier name, quantity, and cost.
- **Read:** View purchase history in an organized table.
- **Update:** Purchase details can be corrected if needed.

- **Delete:** Purchases are cleared during a data wipe.

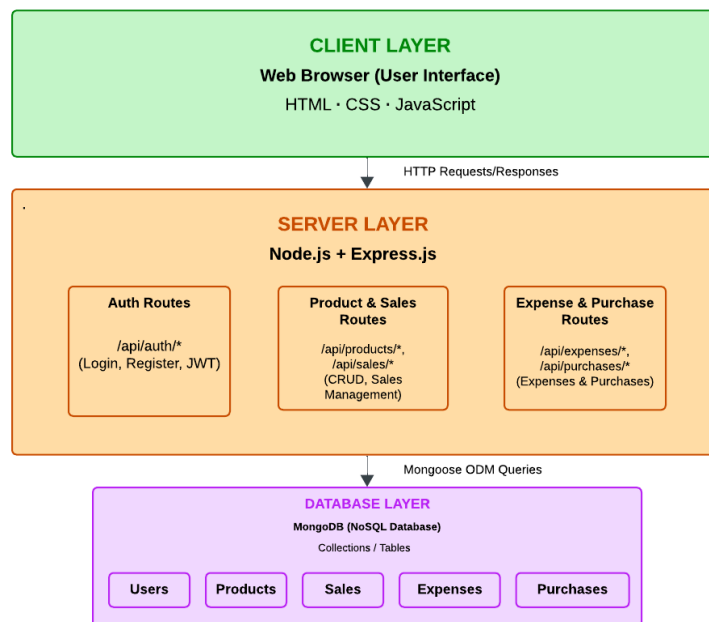
Search and Filtering

The Reports page allows users to filter their financial data by date range. Sellers can choose to view daily, weekly, monthly, or yearly summaries. The Analytics page also has buttons to switch between timeframes, and the bar chart redraws automatically to show the selected period. On the Admin side, the Manage Users page lets admins search through registered users.

User Interface Features

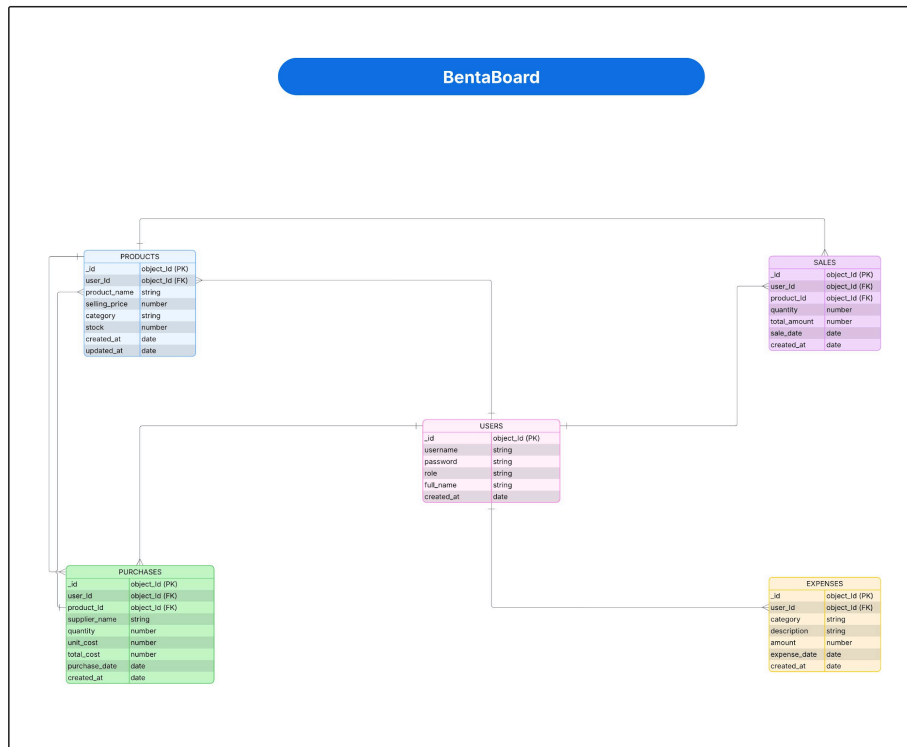
- Dashboard with four summary cards showing Total Sales, Total Expenses, Items Sold, and Net Profit at a glance.
- Responsive design that works on mobile phones, tablets, and laptops.
- Dark mode toggle for users who prefer a darker screen.
- Color-coded buttons: green for adding, blue for editing, red for deleting.
- Striped tables for easier reading.
- Empty state messages that guide users when no data exists yet.
- Loading indicators while data is being fetched.
- Confirmation modals before any delete action.
- Success and error alert messages after every form submission.
- Weather information page using OpenWeatherMap API.

III. Architecture Diagram (Frontend–Backend–Database flow)





IV. Entity Relationship Diagram (ERD)



V. GitHub Link + Commit Logs Summary

- Github Link - [prlanne/final-project-PeNiHAMiNo-BSIT2A](https://github.com/prlanne/final-project-PeNiHAMiNo-BSIT2A)
- Commit Logs Summary:

Commits on May 13, 2026		
Merge pull request #10 from prlanne/Email-verification	Verified	d69ec59
aze-ish authored 9 minutes ago		
feat: Add verification routes and update root redirect to role selection page		b469d47
aze-ish committed 16 minutes ago		
feat: Add Poppins font files for various weights		b8cf6aa
aze-ish committed 23 minutes ago		
feat: Implement PWA support and enhance mobile sidebar functionality		ba60025
aze-ish committed 51 minutes ago		
Updated analytics.html		0030353
aze-ish committed 53 minutes ago		
Updated admin-users.html		c8ac96c
aze-ish committed 54 minutes ago		
Updated admin-settings.html		b945eae
aze-ish committed 54 minutes ago		
Updated admin-reports.html		58fa3aa
aze-ish committed 1 hour ago		
Updated admin-login.html		391ea1a
aze-ish committed 1 hour ago		
Updated admin-dashboard		0f2b343
aze-ish committed 1 hour ago		



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update package json aze-ish committed 1 hour ago	047fd4f		
verif aze-ish committed 1 hour ago	53f8dbf		
userController aze-ish committed 1 hour ago	61177f2		
productController aze-ish committed 1 hour ago	a33e6ce		
authController aze-ish committed 1 hour ago	63c1fb8		
Commits on May 11, 2026			
Updated admin-script.js aze-ish committed 2 days ago	60edeab		
Updated auth.js aze-ish committed 2 days ago	2d9c4e7		
Merge pull request #7 from prlanne/frontend aze-ish authored 2 days ago	706d308		
Clean up inline script aze-ish committed 2 days ago	260ec39		
Remove missing notification CSS link aze-ish committed 2 days ago	83dd697		

Remove missing notification CSS link aze-ish committed 2 days ago	83dd697		
Add PWA meta tags and scripts aze-ish committed 2 days ago	e7892ef		
Add PWA meta tags and manifest link aze-ish committed 2 days ago	0179637		
Add PWA meta tags and manifest link aze-ish committed 2 days ago	60436b2		
Add PWA meta tags and manifest link aze-ish committed 2 days ago	c801756		
Add install notification popup aze-ish committed 2 days ago	51bd97a		
Commits on May 10, 2026			
Added routes for verification endpoints in authentication aze-ish committed 3 days ago	39054cb		
Added controller for handling verification logic in auth flow aze-ish committed 3 days ago	c20dc24		
Added verification code model for email verification during authentication aze-ish committed 3 days ago	d03d0e7		
Updated dependencies for authentication packages aze-ish committed 3 days ago	cbcad21		



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Commits on May 10, 2026		
Added routes for verification endpoints in authentication	aze-ish committed 3 days ago	39054cb
Added controller for handling verification logic in auth flow	aze-ish committed 3 days ago	c20dc24
Added verification code model for email verification during authentication	aze-ish committed 3 days ago	d03d0e7
Updated dependencies for authentication packages	aze-ish committed 3 days ago	cbcad21
Updated server configuration for auth middleware	aze-ish committed 3 days ago	f25447e
Updated User model to support authentication fields	aze-ish committed 3 days ago	c85de4b
Updated user controller for authentication flow	aze-ish committed 3 days ago	257f2e3
Updated main script for authenticated user sessions	prianne committed 3 days ago	ff2cf0a
Updated registration page with authentication features	prianne committed 3 days ago	686aca7
Updated authentication logic and token handling	prianne committed 3 days ago	d8b8905

Commits on May 10, 2026		
Updated styles for authentication UI	prianne committed 3 days ago	fb7ca97
Commits on May 5, 2026		
Add final project lab report for BentaBoard application	prianne committed last week	ee2e536
Commits on May 2, 2026		
Style: update settings view to link with modular settings.js	prianne committed 2 weeks ago	d52b329
Style: update weather integration view to link with modular weatherapi.js	prianne committed 2 weeks ago	b6ef5be
Style: update reporting views to link with modular analytics.js and reports.js	prianne committed 2 weeks ago	1807694
Style: update sales logging view to link with modular saleslogs.js	prianne committed 2 weeks ago	f833b27
Style: update financial views to link with modular expenses.js and purchases.js	prianne committed 2 weeks ago	f167143
Style: update inventory view to link with modular inventory.js	prianne committed 2 weeks ago	b7fc6e8
Style: update dashboard and landing pages to link with modular dashboard.js and hub.js	prianne committed 2 weeks ago	f29dc32



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Style: update role selection view to link with modular role-select.js prlanne committed 2 weeks ago	7cfced2		
Style: update authentication views to link with modular auth.js prlanne committed 2 weeks ago	a97a323		
Refactor: update administrative script logic to align with modular architecture prlanne committed 2 weeks ago	d5da797		
Refactor: migrate data visualization and analytics tracking logic to analytics.js prlanne committed 2 weeks ago	454cd0d		
Refactor: migrate user authentication and session management logic to auth.js prlanne committed 2 weeks ago	030027b		
Refactor: migrate core dashboard UI updates and state management to dashboard.js prlanne committed 2 weeks ago	2a8e2af		
Refactor: migrate expense tracking and financial calculation logic to expenses.js prlanne committed 2 weeks ago	c602004		
Refactor: migrate central navigation and hub connectivity logic to hub.js prlanne committed 2 weeks ago	825dd04		
Refactor: move inventory management and stock logic to inventory.js prlanne committed 2 weeks ago	6539c96		
Refactor: migrate application entry point and initialization to main.js prlanne committed 2 weeks ago	823df4e		
Refactor: extract alert and notification handling from script.js to notifications.js prlanne committed 2 weeks ago	885df0f		

Refactor: extract alert and notification handling from script.js to notifications.js prlanne committed 2 weeks ago	885df0f		
Refactor: migrate purchase transaction and procurement logic to purchases.js prlanne committed 2 weeks ago	55046cb		
Refactor: migrate report generation and export logic to reports.js prlanne committed 2 weeks ago	59bb6f3		
Refactor: move user role selection and access control logic to role-select.js prlanne committed 2 weeks ago	2733848		
Refactor: move sales tracking and transaction logging to saleslogs.js prlanne committed 2 weeks ago	162f7f3		
Refactor: migrate system configuration and user settings logic to settings.js prlanne committed 2 weeks ago	481c818		
Refactor: move common utility functions and shared variables to shared.js prlanne committed 2 weeks ago	d9fa2e2		
Refactor: migrate weather API integration from script.js to weatherAPI.js prlanne committed 2 weeks ago	35e21ff		
Cleanup: delete script.js after modularizing code into specific feature scripts prlanne committed 2 weeks ago	8de616c		

Commits on May 1, 2026		
Updated the API base URL and enhance notification system comments; improve report rendering for better print support	abafa05	 
 prlanne committed 2 weeks ago		
Update print styles for seller reports and enhance layout for better readability	e53e8bf	 
 prlanne committed 2 weeks ago		
Update admin reports layout and enhance print functionality	5469a89	 
 prlanne committed 2 weeks ago		
Merge pull request #5 from prlanne/backend	Verified 796b8ff	 
 prlanne authored 2 weeks ago		
Now searches BOTH User and Admin collections	5948e22	 
 aze-ish committed 2 weeks ago		
Moved dataRouter import and registration BEFORE start() function	d64ebd6	 
 aze-ish committed 2 weeks ago		
Changed to use mongoose.model() with the default connection	0faffe0	 
 aze-ish committed 2 weeks ago		

Commits on May 1, 2026		
Changed to use mongoose.model() with the default connection (same as all other models)	58f055d	 
 aze-ish committed 2 weeks ago		
Commits on Apr 29, 2026		
Add Phase 5 Lab Report: Document data retrieval and dynamic display implementation	60315f5	 
 prlanne committed 2 weeks ago		
Refine Phase 4 Lab Report: Improve formatting, clarity, and document structure	cc4e095	 
 prlanne committed 2 weeks ago		
Updated Phase 3 Lab Report: Improve formatting and clarity of content	46166e0	 
 prlanne committed 2 weeks ago		
Update README.md in backend for improved clarity and organization of models and controllers	c2d1218	 
 prlanne committed 2 weeks ago		
Commits on Apr 28, 2026		
Added Phase 4 Lab Report Documentation	02dc5f7	 
 prlanne committed 2 weeks ago		
Add Admin UI screenshots to lab11 directory	28b0f7a	 
 prlanne committed 2 weeks ago		
Removed Phase 2 and Phase 3 LabReport from frontend and backend folder	48d3dbe	 
 prlanne committed 2 weeks ago		



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Added Phase 4 Lab Report Documentation prlanne committed 2 weeks ago	02dc577		
Add Admin UI screenshots to lab11 directory prlanne committed 2 weeks ago	28b077a		
Removed Phase 2 and Phase 3 LabReport from frontend and backend folder prlanne committed 2 weeks ago	48d3dbe		
Refactor: merge admin-theme.css into style.css prlanne committed 2 weeks ago	519dee4		
Implement features to enhance user experience and optimize performance prlanne committed 2 weeks ago	8be2632		
Allow admin to fetch all from all sellers revenue prlanne committed 2 weeks ago	0d864fe		
Remove version info from role selection page prlanne committed 2 weeks ago	27b3225		
Refactor: clean up admin settings page by removing theme toggle comments and unused stylesheet prlanne committed 2 weeks ago	6b5a963		
Refactor: remove inline admin authentication script and link to external script prlanne committed 2 weeks ago	6878168		
Changed admin dashboard Transactions card to display total revenue from all sellers prlanne committed 2 weeks ago	c6e8ded		

Replace admin Transactions card with All Sellers Total Revenues prlanne committed 2 weeks ago	aaddbad		
Updated Phase 3 Lab Report to include detailed frontend folder structure and admin features prlanne committed 2 weeks ago	f24603b		
delete: remove outdated Phase 3 Lab Report document prlanne committed 2 weeks ago	c174cf6		
Updated Lab PDF to include full-stack Admin view and interface layouts prlanne committed 2 weeks ago	734b09e		
Commits on Apr 27, 2026			
deleted the empty file: Phase 3 Lab Report PDF from documentation prlanne committed 2 weeks ago	c6bf886		
Commits on Apr 26, 2026			
Added Phase 3 Lab Report Documentation prlanne committed 2 weeks ago	96312fd		
Merge pull request #4 from prlanne/frontend prlanne authored 2 weeks ago	443b281		
refactor: update main script for role-based routing and auth guards prlanne committed 2 weeks ago	f641419		
feat: implement core logic for admin dashboard and management prlanne committed 2 weeks ago	011695e		



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feat: implement core logic for admin dashboard and management prlanne committed 2 weeks ago	011695e		
style: update global styles for landing page and glassmorphism components prlanne committed 2 weeks ago	6ed27b3		
feat: add dedicated admin theme for dashboard styling prlanne committed 2 weeks ago	5428d9b		
style: add back-to-role-selection link to registration page prlanne committed 2 weeks ago	a395669		
style: add back-to-role-selection link to login page prlanne committed 2 weeks ago	37dd488		
feat: create role selection landing page for Seller and Admin entry prlanne committed 2 weeks ago	cfbdc49		
feat: implement user management interface for admin control prlanne committed 2 weeks ago	9b642be		
feat: add admin settings page for system configuration prlanne committed 2 weeks ago	7ceb807		
feat: create seller reports view for data monitoring prlanne committed 2 weeks ago	3d6cfd7		
feat: implement secure login portal for administrators prlanne committed 2 weeks ago	9f922c6		
feat: add admin dashboard hub for system overview prlanne committed 2 weeks ago	03c8d68		

Updated dataController aze-ish committed 2 weeks ago	f84de54		
Updated UserRoutes.js: Added protect middleware to GET/PUT/DELETE aze-ish committed 2 weeks ago	a9bba1e		
Updated userController: Admins and users now use separate collections, with login blocking admins from the user endpoint aze-ish committed 2 weeks ago	9356444		
Updated saleController: getSales supports ?userID for admin aze-ish committed 2 weeks ago	2c2c94e		
Updated purchaseController: getPurchases supports ?userID for admin aze-ish committed 2 weeks ago	d875842		
Updated User.js: Now uses test database, added collection: 'users' aze-ish committed 2 weeks ago	23775dc		
Added Admin.js for separate admin interface aze-ish committed 2 weeks ago	36a6989		
Merge pull request #2 from prlanne/backend aze-ish authored 2 weeks ago	Verified 7a0cc4a		
Updated productController to support userID for admin access aze-ish committed 2 weeks ago	5e6fc5c		
Updated expenseController to support userID queries for admin access aze-ish committed 2 weeks ago	807014b		
Updated authController.js: Now uses Admin Model aze-ish committed 2 weeks ago	9d013cd		



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Commits on Apr 25, 2026

Update README.md for Phase 3 completion

prianne committed 3 weeks ago

a0e3a8f  <>

Commits on Apr 22, 2026

Added Phase 3 LabReport/Documentation in the docs folder

prianne committed 3 weeks ago

18f94a7  <>

Cleanup: Remove .gitkeep from frontend folder

prianne committed 3 weeks ago

950ed2e  <>

Added README.md in the frontend folder

prianne committed 3 weeks ago

3f71f0a  <>

Commits on Apr 21, 2026

Added the frontend screenshots to the lab11 folder in docs directory

prianne committed 3 weeks ago

cf22df3  <>

Added script.js to frontend/js folder

prianne committed 3 weeks ago

afffbce  <>

Added bentaboard logo to frontend/img folder

prianne committed 3 weeks ago

aa09994  <>

Added style.css to frontend/css folder

prianne committed 3 weeks ago

7b9a811  <>

Added weather.html to frontend folder

prianne committed 3 weeks ago

b3779ed  <>

Commits on Apr 21, 2026

Added the frontend screenshots to the lab11 folder in docs directory

prianne committed 3 weeks ago

cf22df3  <>

Added script.js to frontend/js folder

prianne committed 3 weeks ago

afffbce  <>

Added bentaboard logo to frontend/img folder

prianne committed 3 weeks ago

aa09994  <>

Added style.css to frontend/css folder

prianne committed 3 weeks ago

7b9a811  <>

Added weather.html to frontend folder

prianne committed 3 weeks ago

b3779ed  <>

Added settings.html to frontend folder

prianne committed 3 weeks ago

fd153f4  <>

Added saleslog.html to frontend folder

prianne committed 3 weeks ago

b42d1b5  <>

Added reports.html to frontend folder

prianne committed 3 weeks ago

239a022  <>

Added register.html to frontend folder

prianne committed 3 weeks ago

7c1728e  <>

Added purchases.html to frontend folder

prianne committed 3 weeks ago

5da01b0  <>



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Added settings.html to frontend folder	fd153f4	
prlanne committed 3 weeks ago		
Added saleslog.html to frontend folder	b42d1b5	
prlanne committed 3 weeks ago		
Added reports.html to frontend folder	239a022	
prlanne committed 3 weeks ago		
Added register.html to frontend folder	7c1728e	
prlanne committed 3 weeks ago		
Added purchases.html to frontend folder	5da01b0	
prlanne committed 3 weeks ago		
Added login.html to frontend folder	6958bbc	
prlanne committed 3 weeks ago		
Added inventory.html to frontend folder	49a308d	
prlanne committed 3 weeks ago		
Added index.html to frontend folder	6aefbd9	
prlanne committed 3 weeks ago		
Added expenses.html to frontend folder	752fff9	
prlanne committed 3 weeks ago		
Added dashboard.html to frontend folder	52aad1d	
prlanne committed 3 weeks ago		
Added analytics.html to frontend folder	77ece47	
prlanne committed 3 weeks ago		

Commits on Apr 20, 2026

Created dataRoutes.js with DELETE API	7973808	
aze-ish committed 3 weeks ago		
Updated changes in folder routes by adding JWT Token and import middleware	9ac23a1	
aze-ish committed 3 weeks ago		
Updated changes inside models folder by adding userID each file	367e975	
aze-ish committed 3 weeks ago		
Updated changes inside middleware folder	8475857	
aze-ish committed 3 weeks ago		

Commits on Apr 20, 2026

Created dataController.js with wipeUserData	d901d71	
aze-ish committed 3 weeks ago		
Updated changes inside controllers folder	08c41c1	
aze-ish committed 3 weeks ago		
Updated Changes with Fixed paths to match the folder	aaebb19	
aze-ish committed 3 weeks ago		

Commits on Apr 13, 2026

Merge branch 'main' of https://github.com/prlanne/final-project-PeNiHAMiNo-BSIT2A	a6c5cc6	
aze-ish committed last month		
Added Phase 2 LabReport /Documentation in the docs folder	82db41e	
aze-ish committed last month		
Merge branch 'main' of https://github.com/prlanne/final-project-PeNiHAMiNo-BSIT2A	4da8857	
prlanne committed on Apr 13		
docs: finalize backend README with tech stack, folder structure, and setup guides	f14ab9c	
prlanne committed on Apr 13		
Added routes	6f62532	
aze-ish committed on Apr 13		
Added models	afa38a2	
aze-ish committed on Apr 13		



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Commits on Apr 13, 2026		
Merge branch 'main' of https://github.com/prlanne/final-project-PeNiHAMiNo-BSIT2A	a6c5cc6	
aze-ish committed last month		
Added Phase 2 LabReport /Documentation in the docs folder	82db41e	
aze-ish committed last month		
Merge branch 'main' of https://github.com/prlanne/final-project-PeNiHAMiNo-BSIT2A	4da8857	
prlanne committed on Apr 13		
docs: finalize backend README with tech stack, folder structure, and setup guides	f14ab9c	
prlanne committed on Apr 13		
Added routes	6f62532	
aze-ish committed on Apr 13		
Added models	afa38a2	
aze-ish committed on Apr 13		
Added middleware and controllers	6a70150	
aze-ish committed on Apr 13		
Add backend configuration files	acee4b2	
aze-ish committed on Apr 13		
Add Express server initialization and middleware setup	4741efa	
aze-ish committed on Apr 13		
Initialized backend environment and configured .gitignore	f9fcbe1	
aze-ish committed on Apr 13		

Commits on Mar 29, 2026		
Added README.md	0bcfbd3	
prlanne authored on Mar 29		
Added planning docs: ERD, UML, System Architecture Diagram and Final Project Proposal	b7ebd5d	
prlanne committed on Mar 29		
Added base folders	4a6f74f	
prlanne committed on Mar 29		

VI. Render Link / Screenshots

- <https://final-project-penihamino-bsit2a-1.onrender.com>

VII. Summary of Division of Tasks

Division of Tasks:

Role	Members	Phase 7 Contributions
Hazel Ann Obidece	Project Manager & Backend Developer	Led the final testing process, verified all API endpoints returned correct responses, fixed backend bugs found during testing, set up the Render deployment configuration, coordinated the team to meet deadlines
Noe Jr Remolar	Documentation Officer	Conducted thorough functional testing on every page, created the bug checklist, tested form validation with valid and invalid inputs, wrote the testing results section, verified that all features worked after deployment
Michael Mantes	Database Manager	Checked data consistency across all collections, verified that database operations correctly updated MongoDB after deployment, ensured schema alignment between code and database, confirmed data wipe function properly cleared all user records
Nica Alyssa Racadag	Frontend Developer	Polished the user interface layout and alignment, fixed responsiveness issues on mobile screens, added loading indicators and empty state messages, finalized dark mode colors and styles, ensured all modals and forms behaved correctly
Pearl Anne Odo	GitHub Manager	Cleaned up the repository by removing unused files and

		old branches, merged all final code into the main branch, wrote the complete README with installation instructions and API documentation, created the v1.0-final release tag, organized the final folder structure
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VIII. Bug testing results / final test notes

IX. Individual reflections from each group member (3–5 sentences)

Hazel Ann Obidece - Project Manager & Backend Developer

This final phase was about making sure everything we built actually works together as one complete system. I focused on testing every API endpoint and fixing the bugs our team found. It was rewarding to see how issues we might have missed in earlier phases were caught and resolved during testing. Deploying to Render was a new experience for me, and it felt amazing to see our project live on the internet. Looking back at where we started from just an idea and some diagrams, I am proud of how far we have come. This project taught me that building software is not just about writing code; it is about patience, teamwork, and paying attention to details.

Noe Jr Remolar - Documentation Officer

My main job in Phase 7 was to test everything thoroughly and document what I found. I went through every page, clicked every button, and tried to break the system by entering wrong information on purpose. Finding bugs was actually satisfying because it meant we could fix them before anyone else saw them. I learned that testing is not just about checking if something works but also about thinking like a user who might do unexpected things. Writing the final documentation helped me understand our entire project better, from the database structure to the user interface. This experience showed me how important documentation is in making a project complete and understandable.

Michael Mantes - Database Manager

In this final phase, I made sure our database was consistent and all the data was being saved and retrieved correctly. After deployment, I checked that the MongoDB collections matched what we designed in our ERD. I also verified that when a user deletes their data, it actually gets removed from the right collections. One thing I learned is that a well-designed database makes everything else easier. When our schemas were correct from the start, we did not have to do major fixes later. This project gave me real experience in managing a database for a working application, and I now understand why planning the data structure early is so important.



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Nica Alyssa Racadag - FrontEnd Manager

As a frontend manager my focus in Phase 7 was making BentaBoard look professional and work smoothly on any device. I spent time adjusting margins, aligning elements, and making sure all the Bootstrap components looked consistent. The dark mode feature was one of my favorite things to work on because it required careful color selection to make sure everything was still readable. I also added loading spinners and empty state messages so users never feel confused about what is happening. Testing on mobile showed me areas where the layout could be better, and I made adjustments to improve the experience on smaller screens. This phase taught me that good design is about making things simple and clear for the user.

Pear Anne Odo - GitHub Manager

My role in this final phase was to organize our repository and make it presentable. I cleaned up old branches, removed files we were not using anymore, and made sure our commit history told a clear story of our development process. Writing the README was a big task because I had to explain our entire project in a way that someone new could understand. I included installation steps, a list of features, and documentation for all our API endpoints. Creating the release tag felt like a milestone because it marked the official completion of our project. This experience taught me how version control and documentation are essential parts of professional software development.

NOTE:

AI Usage: *On creating this Phase 7 Lab Report (documentation), around 20% of AI usage was obtained to guide us on finishing this report such as: asking assistance when encountering errors and other suggestions and recommendations that would serve as our guide for improvement. Overall, we hereby acknowledge that all the information and key ideas presented on this phase was based on our actual work, outputs, and understanding of the concepts.*